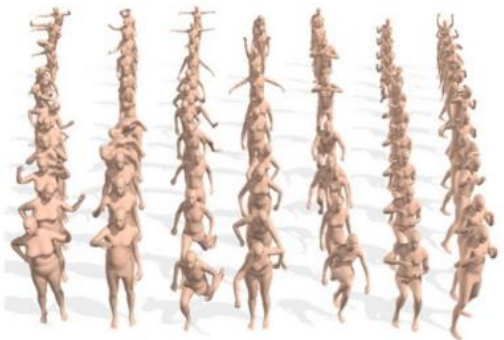


Mocap Dataset

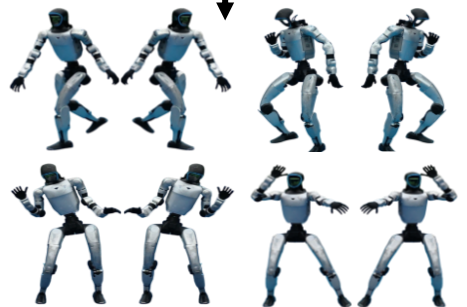


Inverse kinematics
Retargeting

Retargeting Mocap Dataset

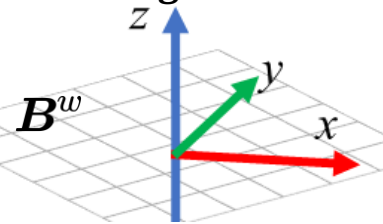


Motion Mirroring



Stubborn

Yaw-Aligned Frame



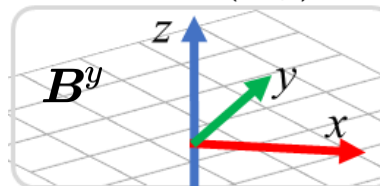
World Frame

Yaw Alignment

$$R_z(-\psi)$$

Remove heading rotation

$$B^y = -R_z(-\psi)B^w$$



Yaw-Aligned Frame

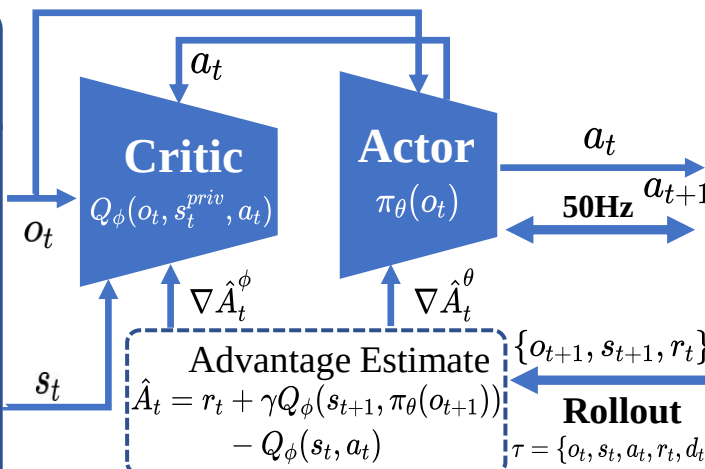
Asymmetric Actor-Critic Training

Yaw-Aligned Frame/
Tracking Command

Target root pose/vel,
joint states, body states

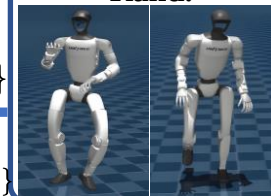
Proprioception State

Privileged Info S_t



IsaacLab
Simulation

- Contacts
- Terrain
- Domain Rand.



Error-Based Adaptive Sampling

$$r_t^{track} = f(s_t; s_t^{ref})$$

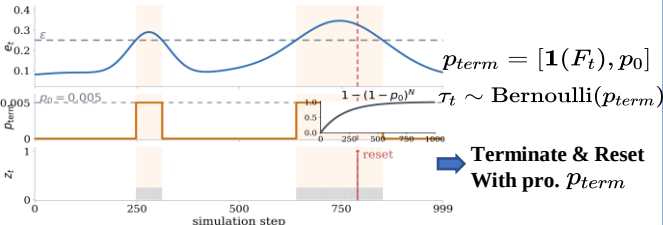


Bin-based Tracking Reward

Sampling Weight

$$w_t = f(r_t^{track})$$

Probabilistic Termination

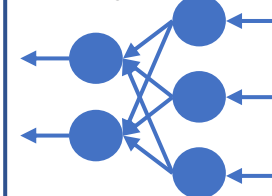


Deployment

Real robot



Policy



Motion
Tracking

